ALGEBRA 2 · TRIGONOMETRIC FUNCTIONS

Unit Circle to Sine Wave

Explore & Reflect

Name _____ Date _____ Period _____

Open **Unit Circle to Sine Wave** on Mr. K's Classroom: Algebra 2 → Trigonometric Functions → Interactive Math. Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3, then do Part 4 on your own.

1 Predict

before you touch the tool

1. At 90°, what is the height of the point on the unit circle?

2. Is sin 210° positive or negative?

2 Explore

use the tool and record what you see

Unit circle values

Drag the point to each angle.

θ (degrees)	θ (radians)	cos θ	sin θ	Quadrant
0°				
30°				
45°				
90°				
150°				
225°				
300°				

Transform the wave $y = A \sin(B(x - C)) + D$

Equation	Amplitude	Period	Phase shift	Midline
$y = 3 \sin x$				
$y = \sin 2x$				
$y = 2 \sin(\frac{1}{2}x) + 1$				
$y = -\sin(x - \frac{\pi}{4})$				

3 Write the rule

fill in each blank from what you observed

1. On the unit circle, cos θ is the _____-coordinate and sin θ is the _____-coordinate.

2. sin θ is positive in Quadrants _____. cos θ is positive in Quadrants _____.

3. For $y = A \sin(B(x - C)) + D$, the amplitude is _____ and the period is _____.

4. D moves the midline _____. C shifts the wave _____.

4 Check no tool: show your work

1. Find cos 120° exactly.

2. Find $\sin \frac{7\pi}{6}$ exactly.

3. Find the period of $y = 4 \sin 3x$.

4. Find the amplitude and midline of $y = -5 \sin x + 2$.

5 Reflect

Why does the sine graph repeat every 2π?